EECS 545 Homework 5

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1 K-means and GMM for Image Compression

(b)

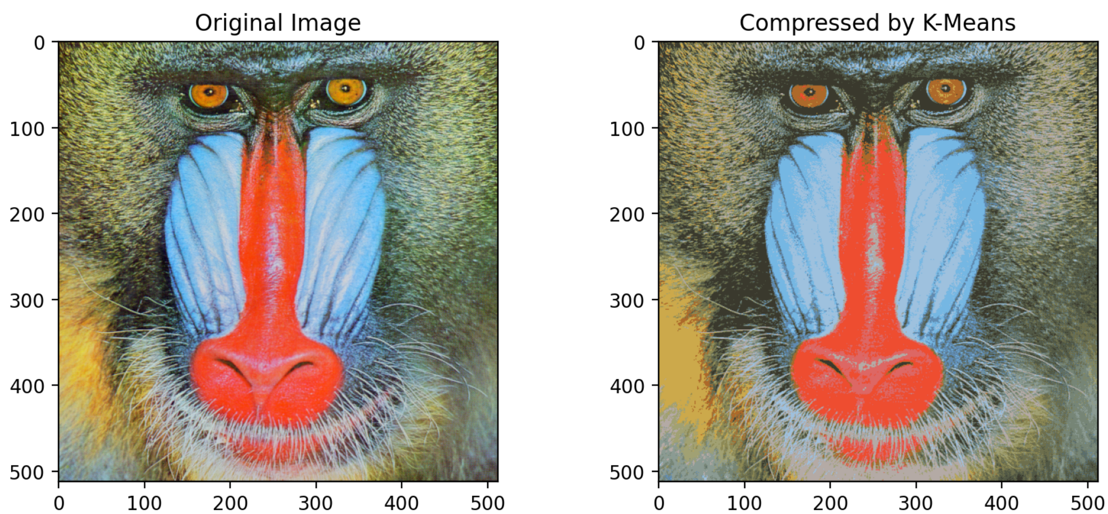

Figure 1: K-means compression ($K = 16$)

The mean pixel error is 14.61.

(e) A GMM with $K = 5$ full-covariance components was trained via EM for 50 iterations on `mandrill-small.tiff`. The final log-likelihood converged to -228103.43.

k	π_k	μ_k (R, G, B)
1	0.13	(180.22, 149.48, 116.78)
2	0.43	(125.08, 136.46, 114.67)
3	0.11	(233.67, 86.85, 67.54)
4	0.21	(81.61, 89.01, 69.69)
5	0.11	(140.02, 189.46, 225.51)

(f) Compressing `mandrill-large.tiff` via MAP assignment under the GMM gives a mean pixel error of 16.37.

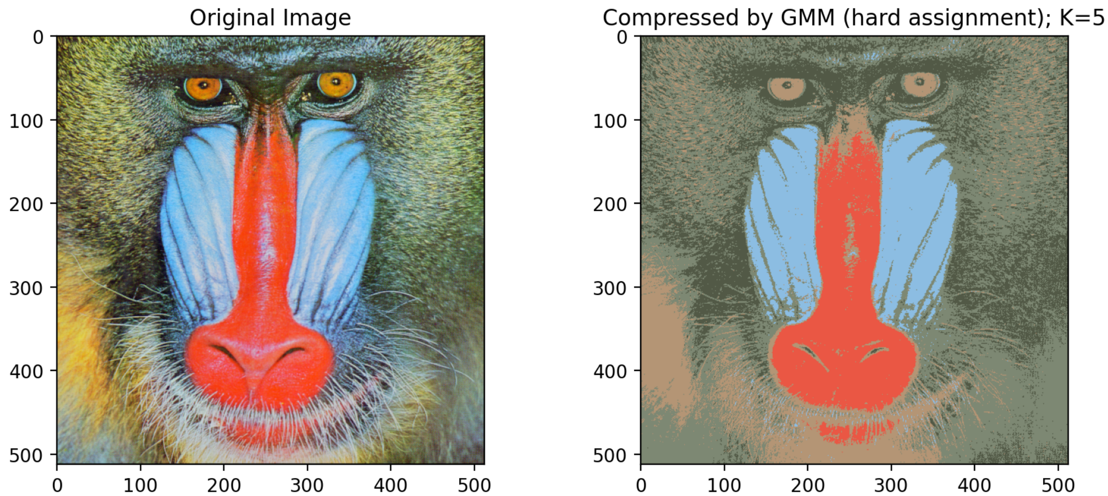

Figure 2: GMM compression ($K = 5$)

2 EM for GDA with Missing Labels

(a) The labeled part of $\mathcal{J}$ is already a sum of log-probabilities so it stays as-is. For each unlabeled point $\mathbf{x}^{(i)}$, we introduce an arbitrary distribution $Q_{ij} = q_i(y^{(i)} = j)$ with $Q_{i0} + Q_{i1} = 1$ and apply Jensen's inequality to the concave log

$$\log \sum_j p(\mathbf{x}^{(i)}, y^{(i)} = j) = \log \sum_j Q_{ij} \frac{p(\mathbf{x}^{(i)}, y^{(i)} = j)}{Q_{ij}} \geq \sum_j Q_{ij} \log \frac{p(\mathbf{x}^{(i)}, y^{(i)} = j)}{Q_{ij}}.$$

Substituting back into $\mathcal{J}$ gives the lower bound

$$\mathcal{L}(\mu, \Sigma, \phi) = \sum_{i=1}^{l} \log p(\mathbf{x}^{(i)}, y^{(i)}) + \lambda \sum_{i=l+1}^{l+u} \sum_{j \in \{0,1\}} Q_{ij} \log \frac{p(\mathbf{x}^{(i)}, y^{(i)} = j)}{Q_{ij}}. \quad \blacksquare$$

(b) The bound is tightest when Jensen's inequality holds with equality, which happens when the ratio inside the log is constant in j. That forces $Q_{ij} \propto p(\mathbf{x}^{(i)}, y^{(i)} = j)$, and after normalizing we get

$$Q_{ij} = \frac{p(\mathbf{x}^{(i)}, y^{(i)} = j)}{\sum_{k \in \{0,1\}} p(\mathbf{x}^{(i)}, y^{(i)} = k)} = p(y^{(i)} = j \mid \mathbf{x}^{(i)}) = \frac{\phi_j \, \mathcal{N}(\mathbf{x}^{(i)}; \mu_j, \Sigma_j)}{\sum_k \phi_k \, \mathcal{N}(\mathbf{x}^{(i)}; \mu_k, \Sigma_k)}.$$

(c) Holding Q_{ij} fixed, we maximize $\mathcal{L}$ w.r.t. μ_k. Only the Gaussian terms depend on μ_k, and from labeled data where $y^{(i)} = k$ we get the usual $-\frac{1}{2}(\mathbf{x}^{(i)} - \mu_k)^\top \Sigma_k^{-1} (\mathbf{x}^{(i)} - \mu_k)$ terms while from unlabeled data the same expression appears weighted by λQ_{ik}. Taking the derivative, setting it to zero, and cancelling Σ_k^{-1} yields

$$\mu_k = \frac{\displaystyle\sum_{i=1}^{l} \mathbb{1}[y^{(i)} = k] \, \mathbf{x}^{(i)} + \lambda \sum_{i=l+1}^{l+u} Q_{ik} \, \mathbf{x}^{(i)}}{\displaystyle\sum_{i=1}^{l} \mathbb{1}[y^{(i)} = k] + \lambda \sum_{i=l+1}^{l+u} Q_{ik}}.$$

Intuitively this is the weighted average of all data attributed to class k, where labeled points contribute hard counts ($\mathbb{1}[y^{(i)}=k]$) and unlabeled points contribute soft pseudo-counts ($\lambda\, Q_{ik}$).

(d) The terms in $\mathcal{L}$ involving ϕ are $\log \phi_j = \mathbb{1}[j=1]\log\phi + \mathbb{1}[j=0]\log(1-\phi)$, appearing in both labeled and unlabeled sums. Collecting everything and differentiating w.r.t. ϕ gives

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{\sum_{i=1}^{l} \mathbb{1}[y^{(i)}=1] + \lambda \sum_{i=l+1}^{l+u} Q_{i1}}{\phi} - \frac{\sum_{i=1}^{l} \mathbb{1}[y^{(i)}=0] + \lambda \sum_{i=l+1}^{l+u} Q_{i0}}{1-\phi} = 0$$

$$\implies \phi = \frac{\displaystyle\sum_{i=1}^{l} \mathbb{1}[y^{(i)}=1] + \lambda \sum_{i=l+1}^{l+u} Q_{i1}}{l + \lambda\, u}.$$

Intuitively ϕ is the fraction of effective class-1 counts over all effective counts, where labeled points cast hard votes and unlabeled points cast soft votes weighted by $\lambda\, Q_{i1}$.

(e) By analogy with the M-step for μ_k and the standard GDA/GMM covariance update, the M-step for Σ_k is

$$\Sigma_k = \frac{\displaystyle\sum_{i=1}^{l} \mathbb{1}[y^{(i)}=k]\,(\mathbf{x}^{(i)} - \mu_k)(\mathbf{x}^{(i)} - \mu_k)^\top + \lambda \sum_{i=l+1}^{l+u} Q_{ik}\,(\mathbf{x}^{(i)} - \mu_k)(\mathbf{x}^{(i)} - \mu_k)^\top}{\displaystyle\sum_{i=1}^{l} \mathbb{1}[y^{(i)}=k] + \lambda \sum_{i=l+1}^{l+u} Q_{ik}}.$$

This is the weighted sample covariance around μ_k with hard counts for labeled data and soft pseudo-counts $\lambda\, Q_{ik}$ for unlabeled data.

3 PCA and Eigenfaces

(a) We can assume WLOG that $\bar{\mathbf{x}} = \mathbf{0}$ (zero-centered data). Let $\mathbf{X} \in \mathbb{R}^{D\times N}$ have the data points as columns so that $\mathbf{S} = \frac{1}{N}\mathbf{X}\mathbf{X}^\top$.

Following Hint 2, we rewrite $\mathcal{J} = \frac{1}{N}\|\mathbf{X} - \mathbf{U}\mathbf{U}^\top\mathbf{X}\|_F^2$ and expand using $\|A\|_F^2 = \operatorname{tr}(A^\top A)$

$$\|\mathbf{X} - \mathbf{U}\mathbf{U}^\top\mathbf{X}\|_F^2 = \operatorname{tr}\left(\mathbf{X}^\top\mathbf{X} - 2\,\mathbf{X}^\top\mathbf{U}\mathbf{U}^\top\mathbf{X} + \mathbf{X}^\top\mathbf{U}\,\underbrace{\mathbf{U}^\top\mathbf{U}}_{\mathbf{I}_K}\,\mathbf{U}^\top\mathbf{X}\right) = \operatorname{tr}(\mathbf{X}^\top\mathbf{X}) - \operatorname{tr}(\mathbf{X}^\top\mathbf{U}\mathbf{U}^\top\mathbf{X}).$$

The second trace can be cycled as $\operatorname{tr}(\mathbf{X}^\top\mathbf{U}\mathbf{U}^\top\mathbf{X}) = \operatorname{tr}(\mathbf{U}^\top\mathbf{X}\mathbf{X}^\top\mathbf{U})$, and after dividing by N and recognizing $\mathbf{S}$ we get

$$\mathcal{J} = \operatorname{tr}(\mathbf{S}) - \operatorname{tr}(\mathbf{U}^\top\mathbf{S}\,\mathbf{U}).$$

Since eigenvalues sum to the trace we have $\operatorname{tr}(\mathbf{S}) = \sum_{i=1}^{D}\lambda_i$, and expanding $\mathbf{U} = [\mathbf{u}_1 \cdots \mathbf{u}_K]$ column-by-column gives $\operatorname{tr}(\mathbf{U}^\top\mathbf{S}\,\mathbf{U}) = \sum_{i=1}^{K}\mathbf{u}_i^\top\mathbf{S}\,\mathbf{u}_i$, so $\mathcal{J} = \sum_{i=1}^{D}\lambda_i - \sum_{i=1}^{K}\mathbf{u}_i^\top\mathbf{S}\,\mathbf{u}_i$ as claimed. ■

The first sum is a constant we cannot control, so minimizing $\mathcal{J}$ amounts to making $\sum_{i=1}^{K}\mathbf{u}_i^\top\mathbf{S}\,\mathbf{u}_i$ as large as possible. By the Rayleigh–Ritz theorem this is maximized exactly when each $\mathbf{u}_i$ is the eigenvector for the i-th largest eigenvalue of $\mathbf{S}$, which gives $\mathbf{u}_i^\top\mathbf{S}\,\mathbf{u}_i = \lambda_i$. The minimum distortion is then $\sum_{i=1}^{D}\lambda_i - \sum_{i=1}^{K}\lambda_i = \sum_{k=K+1}^{D}\lambda_k$. ■

(c) The first 10 eigenvalues (sorted in descending order) and a plot of all eigenvalues with log-scale y-axis are shown below.

Index	Eigenvalue
1	2,718,207.41
2	2,610,783.99
3	365,363.88
4	211,062.37
5	110,557.41
6	104,672.42
7	78,480.41
8	67,004.29
9	52,571.02
10	49,176.70

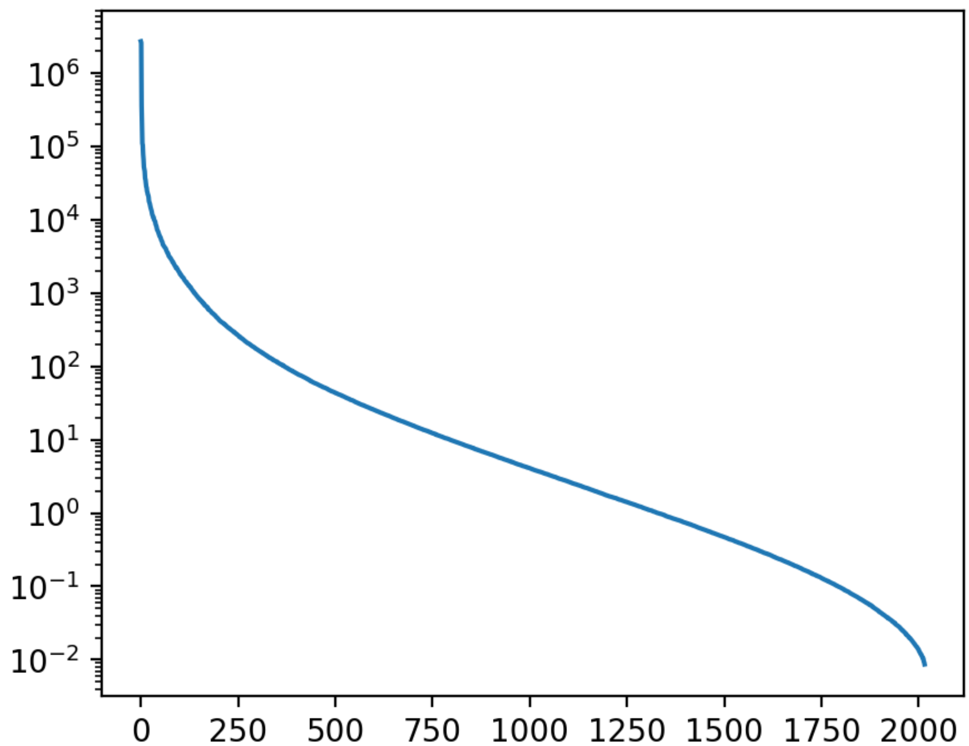

Figure 3: Eigenvalue spectrum (log scale)

(d) The 2×5 grid below shows the mean face (top-left) followed by the first 9 eigenfaces.

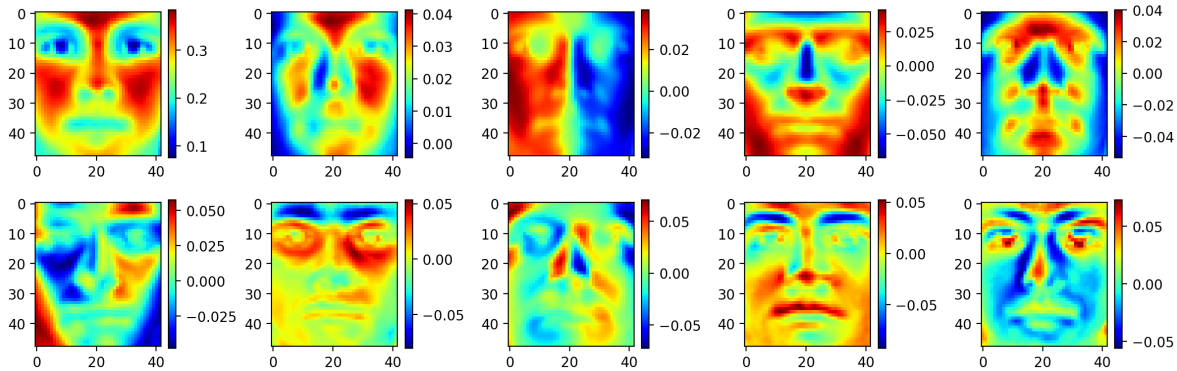

Figure 4: Mean face and first 9 eigenfaces

PC 1 captures overall brightness and illumination level. PC 2 captures left–right lighting asymmetry. PC 3 picks up vertical variation related to overhead lighting or the presence of hair. PCs 4–5 start to encode facial structure like jawline and nose shape. PCs 6–9 capture finer details such as glasses, mouth shape, and eyebrows.

(e) The total dimensionality is $L = 48 \times 42 = 2016$. To preserve 95% of the total variance we need $K = 43$ components (97.87% dimension reduction), and for 99% we need $K = 167$ (91.72% reduction).

4 Independent Component Analysis

(b) The learned 5×5 unmixing matrix W is

$$W = \begin{pmatrix} 72.151 & 28.624 & 25.910 & -17.232 & -21.191 \\ 13.459 & 31.944 & -4.030 & -24.010 & 11.899 \\ 18.897 & -7.804 & 28.715 & 18.144 & -21.175 \\ -6.012 & -4.157 & -1.017 & 13.873 & -5.263 \\ -8.741 & 22.558 & 9.613 & 14.736 & 45.288 \end{pmatrix}.$$

5 Conditional Variational Autoencoders

(a) We introduce an approximate posterior $q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{y})$ and write the marginal as an expectation

$$\log p_\theta(\mathbf{x} \mid \mathbf{y}) = \log \int p_\theta(\mathbf{x}, \mathbf{z} \mid \mathbf{y})\, d\mathbf{z} = \log \mathbb{E}_{q_\phi}\left[\frac{p_\theta(\mathbf{x}, \mathbf{z} \mid \mathbf{y})}{q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{y})}\right].$$

Since log is concave, Jensen's inequality gives us $\geq \mathbb{E}_{q_\phi}\big[\log \frac{p_\theta(\mathbf{x},\mathbf{z}|\mathbf{y})}{q_\phi(\mathbf{z}|\mathbf{x},\mathbf{y})}\big]$. We then factor $p_\theta(\mathbf{x}, \mathbf{z} \mid \mathbf{y}) = p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{y})\, p_\theta(\mathbf{z} \mid \mathbf{y})$ and split the log to obtain

$$\log p_\theta(\mathbf{x} \mid \mathbf{y}) \geq \underbrace{\mathbb{E}_{q_\phi}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{y})\big] - D_{KL}\big(q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{y}) \| p_\theta(\mathbf{z} \mid \mathbf{y})\big)}_{\mathcal{L}(\theta, \phi; \mathbf{x}, \mathbf{y})}. \quad \blacksquare$$

(b) By conditional independence (Eq. 16) the KL splits into m univariate terms. For a single component $q = \mathcal{N}(\mu, \sigma^2)$ vs $p = \mathcal{N}(0, 1)$ we have $D_{KL}(q \| p) = \mathbb{E}_q[\log q] - \mathbb{E}_q[\log p]$. Working these out, $\mathbb{E}_q[\log q] = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{1}{2}$ since the quadratic averages to σ^2, and $\mathbb{E}_q[\log p] = -\frac{1}{2}\log(2\pi) - \frac{\sigma^2+\mu^2}{2}$ using $\mathbb{E}_q[z^2] = \sigma^2 + \mu^2$. The difference simplifies to $\frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log\sigma^2)$, and summing over all m components gives

$$D_{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{y}) \| p_\theta(\mathbf{z} \mid \mathbf{y})) = -\frac{1}{2}\sum_{j=1}^{m}(1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2). \quad \blacksquare$$

(c)

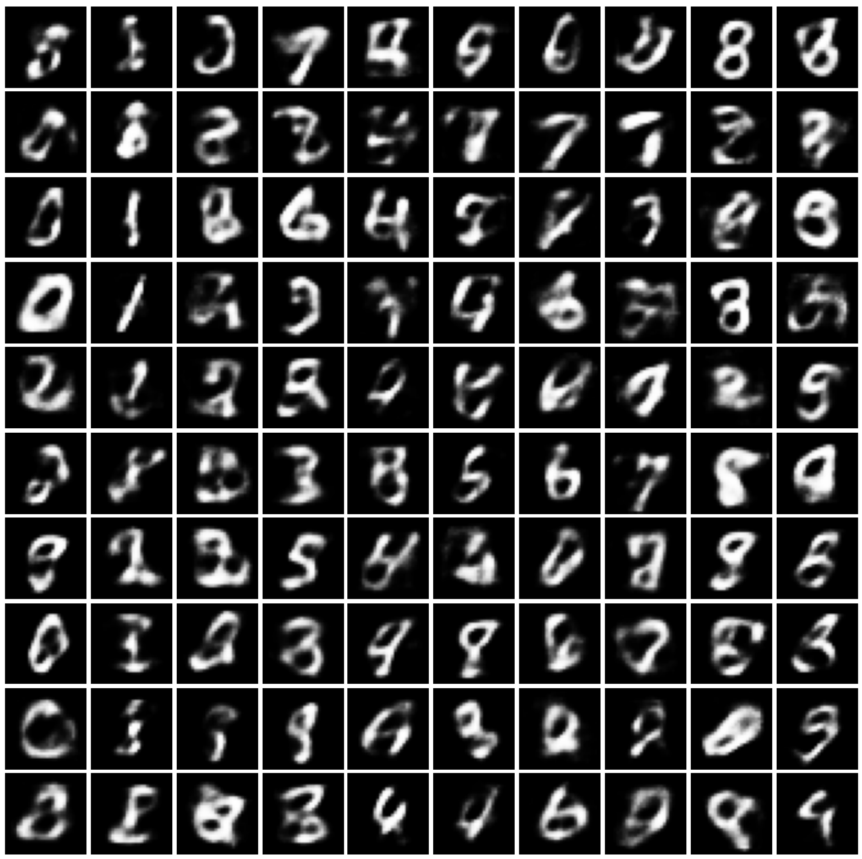

Figure 5: CVAE generated digits (10×10)